

Lab2 Report

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Propose

Collatz Conjecture's proposition is:

For a given sequence,

$$a_n = \begin{cases} \frac{a_{n-1}}{2}, & a_{n-1} \equiv 0 \pmod{2} \\ 3a_{n-1} + 1, & a_{n-1} \equiv 1 \pmod{2} \end{cases}$$

The conjecture is:

$$\forall a_0 \in \mathbb{N}^*, \exists k \in \mathbb{N}^*, s.t. a_k = 1$$

The purpose is to find the minimum k for a small number a_0 which is stored in x3100. Then save k in x3101. The program is processed in LC-3 system, and it is designed with assembly codes.

Principles

There are some tasks to be solved:

1. To judge whether a_n is 1;
2. To judge whether a_n is even or odd;
3. To realize $3 * a_{n-1}$;
4. To realize $\frac{a_{n-1}}{2}$.

To find whether $a_n = 1$, we can judge whether $a_n + (-1) = 0$. The assembly code is:

```
ADD R2,R0,x-1
BRZ OVER
```

If a_n is even, the last bit of it is 0; if a_n is odd, the last bit is 1. So we can judge whether $a_n \text{ AND } 1 = 0$. The assembly code is:

```
AND R2,R0,x1
BRZ EVEN
```

Multiplication is easy to realize by ADD several times. The assembly code to realize $R0 = R0*3+1$ is:

```
ADD R2,R0,R0
;R2 = 2*R0
ADD R0,R2,R0
;R0 = 3*R0
ADD R0,R0,x1
;R0 = 3*R0 + 1
```

Division can be realized by doing subtraction and counting times it needs. To realize $R0 = R0/2$ ($R0$ stores an even number), we can repeat $R0 = R0-2$ until $R0=0$, and record times we repeated. The assembly code is:

```
LOOP:
    ADD R2, R2, x1
    ADD R0, R0, x-2
    ; R0=R0-2, R2++
    BRP LOOP
    ; R2=R0/2
    ADD R0, R2, x0
```

Procedure

The bugs encountered and solutions are as below:

Bug1: The division loop cannot stop.

Solution: BRP should be put just after ADD R0,R0,x-2, but not ADD R2,R2,x1.

Bug1: The data in R1 (k) cannot be stored in x3101 correctly.

Solution: When using PC-relative way (ST) to store, Pcoffset9 should be x3101 - (PC+1).

It's easier to store by the indirect way (STI).

Result

1. x3100: 12
x3101: 9
2. x3100: 14
x3101: 17
3. x3100: 34
x3101: 13
4. x3100: 56
x3101: 19
5. x3100: 32
x3101: 5

All results are correct.